

MARS

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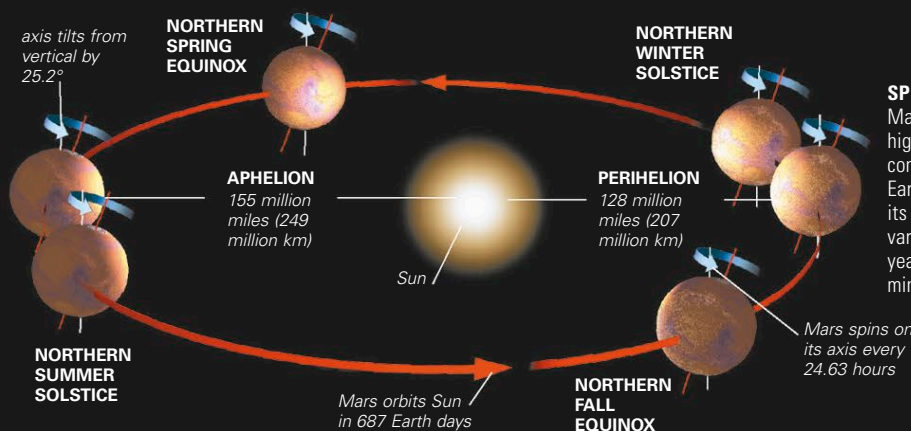
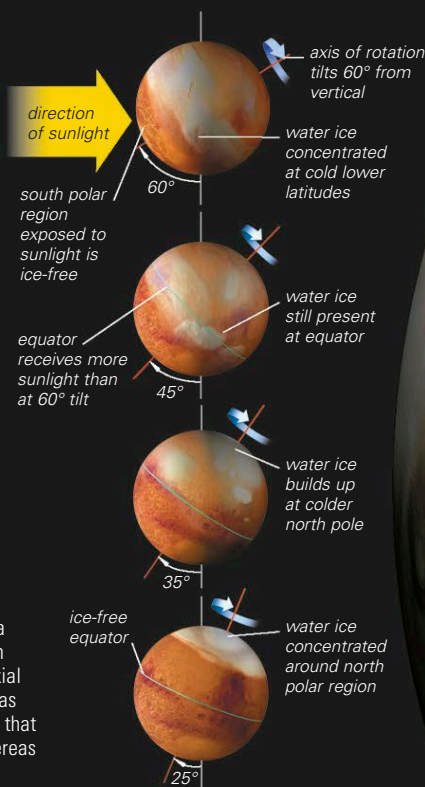
MARS IS THE OUTERMOST of the four rocky planets. Also known as the Red Planet because of its rust-red color, it is named after the Roman god of war. Its varied surface features include deep canyons and the highest volcanoes in the solar system. Although Mars is now a dry planet, a large body of evidence indicates that liquid water once flowed across its surface.

ORBIT

Mars has an elliptical orbit, so at its closest approach to the Sun (perihelion), it receives 45 percent more solar radiation than at the farthest point (aphelion). This means that the surface temperature can vary from -195°F (-125°C) at the winter pole to 77°F (25°C) during the summer. At 25.2° , the current axial tilt of Mars is similar to that of Earth and, like Earth, Mars experiences changes in seasons as the north pole, and then the south pole, points toward the Sun during the course of its orbit. Throughout its history, Mars's axial tilt has fluctuated greatly due to various factors, including Jupiter's gravitational pull. These fluctuations have caused significant changes in climate. When Mars is heavily tilted, the poles are more exposed to the Sun, causing water ice to vaporize and build up around the colder lower latitudes. At a lesser tilt, water ice becomes concentrated at the colder poles.

CHANGES IN AXIAL TILT

Water-ice distribution during a Martian winter in the northern hemisphere varies with the axial tilt. The translucent white areas shown here represent thin ice that melts during the summer, whereas the thick white ice remains.

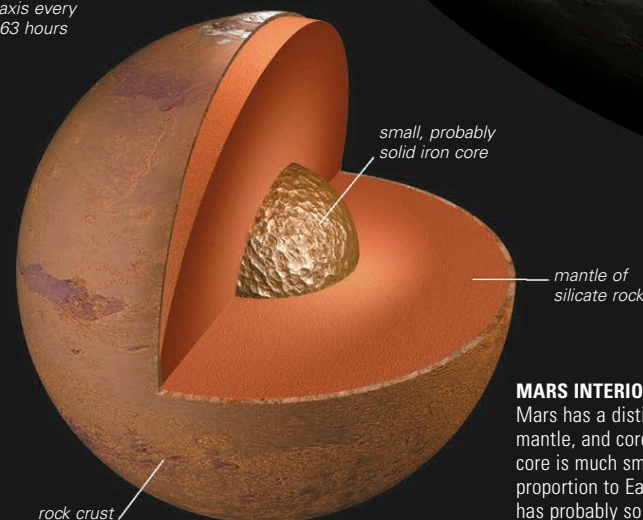


SPIN AND ORBIT

Mars's orbit is highly eccentric compared to that of Earth, which means that its distance from the Sun varies more during a Martian year. A Martian day is 42 minutes longer than an Earth day.

STRUCTURE

Mars is a small planet, about half the size of Earth, and farther away from the Sun. Its size and distance mean that it has cooled more rapidly than Earth, and its once-molten iron core is probably now solid. Its relatively low density compared to the other terrestrial planets indicates that the core may also contain a lighter element, such as sulfur, in the form of iron sulfide. The small core is surrounded by a thick mantle composed of solid silicate rock. The mantle was a source of volcanic activity in the past, but it is now inert. Data gathered by the Mars Global Surveyor spacecraft has revealed that the rocky crust is about 50 miles (80 km) thick in the southern hemisphere, whereas it is only about 22 miles (35 km) thick in the northern hemisphere. Mars has the same total land area as Earth, since it has no liquid water on its surface.



MARS INTERIOR

Mars has a distinct crust, mantle, and core. The core is much smaller in proportion to Earth's and has probably solidified.